# BASIC 6

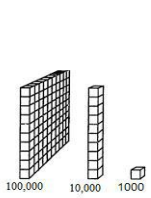
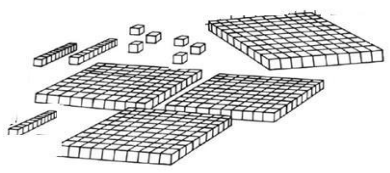
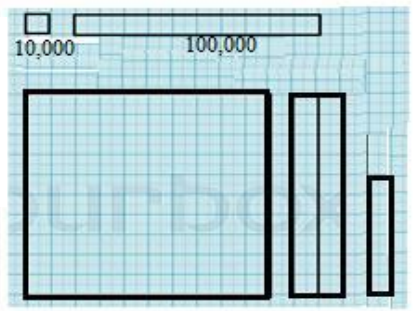
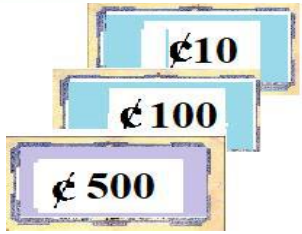


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## Basic 6

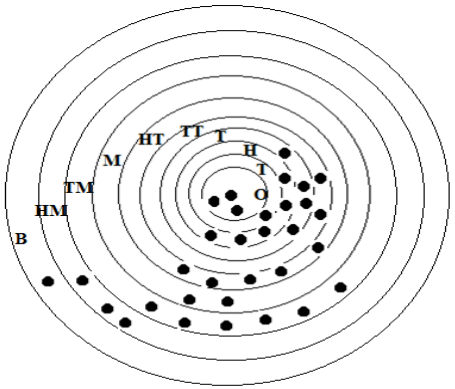
## Strand 1: Number

## Sub-Strand 1: Counting, Representation, Cardinality & Ordinality

| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B6.1.1.1**<br><br>**Demonstrate an understanding of quantities and place value for multi-digit numerals up to 1,000, 000,000 or 1 billion** | **B6.1.1.1.1 Model number quantities up to 1,000,000 using graph sheets and multi-base block**<br><br>E.g.1. Ask pupils to model number quantities up to 1000,000 using graph sheets or multi-base materials. For instance, with multi-base block, a cube = 1000 unit; a rod = 10,000; a flat = 100,000 and a block = 1,000,000; learners model 436,000 with the appropriate materials<br>E.g.2 Ask pupils to model the number 137,000 shading graph sheet square as shown below<br><br>  <br>  <br>  <br><br>E.g.3. Give learners teacher-make token currency notes [¢10, ¢100 & ¢500 notes] on different coloured-paper and ask them work out how many will be required to model given amounts up to ¢1,000,000; e.g. ¢123,480.<br><br>  | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative learning; Personal Development and Leadership; Attention to Precision |



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| **B6.1.1.1**<br>**Demonstrate an understanding of quantities and place value for multi-digit numerals up to 1,000, 000,000 or 1 billion CONT'D** | **B6.1.1.1.2 Read and write numbers in figures and in words up to 1,000,000,000**<br>E.g. 1 Play the place value number wheel game: Use the million number wheel to generate 7-digit numbers and represent the number generated on a place value frame. Throw a number of pebbles (or stones) onto the number wheel, identify the number of pebbles landing in each place value ring, and state the number generated in the wheel (i.e. 5 landed in the hundred-thousands' ring making the number twenty-thousand or 500,000).<br>NB: The representations to use also include verbal, and numerals. Learners must match number word cards to the figures<br>  <br><br>Place value number wheel<br>Place value chart<br>Billion \| Hundred Million \| Ten Million \| Million \| Hundred-thousands \| Ten-thousands \| Thousand \| Hundreds \| Tens \| Ones<br>1 \| 3 \| 6 \| 2 \| 5 \| 2 \| 4 \| 5 \| 1 \| 3<br>E.g. 2. Ask pupils to write given numbers on the expanded form of the number : 1,362,524,513 = 1,000,000,000 × 1 + 100,000,000 × 3 +10,000,000× 6 1,000,000 × 2 + 100,000 × 5 +10,000× 2 + 1,000 × 4 +100 × 5 + 10 × 1 + 1× 3 = 1,000,000,000 + 300,000,000 + 60,000,000 + 2,000,000 + 500,000 + 20,000 + 4000 + 500 + 10 + 3 | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative learning; Personal Development and Leadership Attention to Precision |



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<tr><th>CONTENT STANDARD</th><th>INDICATORS AND EXEMPLARS</th><th>SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES</th></tr>
<tr><td>B6.1.1.1<br><br>Demonstrate an understanding of quantities and place value for multi-digit numerals up to 1,000, 000,000 or 1 billion CONT'D</td><td>B6.1.1.1.3 Identify numbers in different positions around a given number in a number chart<br><br>E.g. 1 Display a number chart in multiples of 1,500 between 10,000 and 50,000 and lead learners identify numbers in different positions around a given number. Put learners in pairs and give each group a number grid and have them identify numbers in different positions around a chosen number<br><br><table><tr><td>10,000</td><td>11,500</td><td>12,000</td><td>13,500</td><td>15,000</td><td>16,500</td></tr><tr><td>20,000</td><td>21,500</td><td>22,000</td><td>23,500</td><td>24,000</td><td>25,500</td></tr><tr><td>30,000</td><td>31,500</td><td>33,000</td><td>34,500</td><td>36,000</td><td>37,500</td></tr><tr><td>40,000</td><td>41,500</td><td>43,000</td><td>44,500</td><td>46,000</td><td>47,500</td></tr><tr><td>50,000</td><td>51,500</td><td>53,000</td><td>54,500</td><td>56,000</td><td>57,500</td></tr><tr><td>60,000</td><td>61,500</td><td>63,000</td><td>64,500</td><td>66,000</td><td>67,500</td></tr></table></td><td>Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership, Attention to Precision</td></tr>
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| **B6.1.1.1**<br><br>**Demonstrate understanding of quantities and place value for multi-digit numerals up to 1,000, 000,000 or 1 billion**<br>**CONT'D** | **B6.1.1.1.4 Compare and order whole numbers up to 100,000 and represent the comparison using ">, <, or ="**<br><br>E.g. 1 Learners identify numbers which are 10,000 more or 10,000 less than a given six-digit number. E.g. 123,400 is 10,000 less than133,400 Learners use words such as: "equal to" "greater than" "less than" and later use symbols such as "<", "=", ">" to compare numbers up to 10,000 taking into consideration the place value of each digit in the given number. E.g. $100200 = 100200, \quad 2746794 > 2646796$<br><br>E.g. 2 Learners work together in their groups to order a given set of numbers in ascending or descending order verbally and in writing. E.g. For instance; $140230, 17025,75\ \ 267389, 287368,$<br><br>E.g. 3 Give two numbers between 10,000 and 100,000 to each group and encourage learners to say as many things as possible about the two numbers. For instance 234675 and 253874; 234675 is less than (smaller than) than 253874 or 253874 is bigger than (greater than) 234675, or 234675 is almost 20000 less than 253874 etc. | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership. Attention to Precision |



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<tr><td>B6.1.1.1<br><br>Demonstrate an understanding of quantities and place value for multi-digit numerals up to 1,000, 000,000 or 1 billion CONT'D</td><td>B6.1.1.1.5 Round (off, up, down) whole numbers up to 100,000 to the nearest ten thousands, thousands, hundreds and tens<br><br>E.g. 1 Learners round off numbers to the nearest 100,000, 10,000 and 1000. For instance; 129,500 is approximately 130,000 and 19100 as approximately 19,000<br><br>E.g. 2 Learners round up and round down to estimate sums and differences.<br><br>NB: To round down learners consider the smaller of the two approximated numbers, while round up considers the bigger of the two numbers. The table below may bring the meaning of these two concepts clearer<br><br><table><tr><th>214765</th><td>Round up</td><td>Round down</td><td>Round off</td></tr><tr><td>to the nearest ten</td><td>214770</td><td>214760</td><td>214970</td></tr><tr><td>to the nearest hundred</td><td>214800</td><td>214700</td><td>214800</td></tr><tr><td>to the nearest thousand</td><td>215000</td><td>214000</td><td>215000</td></tr></table><br>E.g. 3 Give two numbers between 10,000 and 1000,000 to each group and encourage learners to compare and say as many things as possible about the two numbers. For instance 234675 and 253874; 234675 is less than (smaller than) than 253874 or 253874 is bigger than (greater than) 234675, or 234675 is almost 20000 less than 253874 etc.<br><br>E.g. 4 Ask learners to work together to round up and round down to estimate sums and differences.</td><td>Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative learning; Personal Development and Leadership to Precision</td></tr>
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| **B6.1.1.1**<br>**Demonstrate understanding of quantities and place value for multi-digit numerals up to 1,000, 000,000 or 1 billion**<br>**CONT'D** | **B6.1.1.1.6 Skip count forwards and backwards in 5000s, 10,000s etc. up to and from 1,000,000.**<br>E.g. 1 Put learners into convenient groups. A learner mentions a number and another makes skip count in 5000s or 10000s to include the fifth count. For instance.<br>Learner 1: Shout out "287940"<br>Learner 2: 292940, 297940, 312940, 317940, etc.<br>Learner 3; Shout out "827685"<br>Learner 4; 817685, 807685, 797685, 787685, 777685….etc.<br>E.g .2 Skip count forwards and backwards by 5000s and between 10000 and 100000, but make an error or leave out a number. Challenge learners to identify or correct error | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |
| **B6.1.1.2 Demonstrate understanding of Roman Numerals up to C**<br>**(i.e. 100)** | **B6.1.1.2.1 Recognise Roman Numerals system up to C (i.e. 100)**<br>E.g. 1 Display roman numeral charts (1-100) arranged in sequential order and lead learners to identify the numerals. Learners identify the main characters of the roman numerals used to build the table up to 30 i.e. I, II, III, IV, V, X, L and C<br>Call out a numeral and have learners point at it from the chart<br>E.g. 2 Have learners match the Roman numerals to the Hindu-Arabic numerals for instance I = 1; V= 5: IX= 9, XV= 15, XXX = 30, XL = 40, LVI = 56, XCIX = 99. Mention some numerals randomly and have learners point at it on the chart. | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |

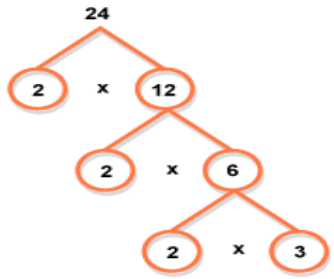
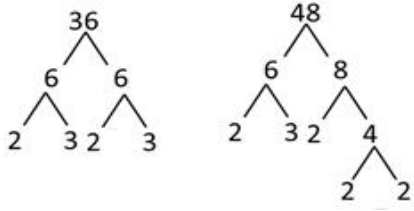
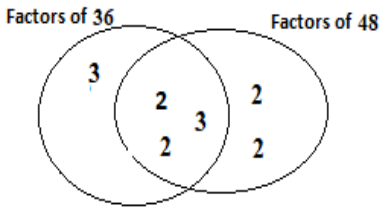


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| **B6.1.1.2 Demonstrate understanding of Roman Numerals up to C**<br>**(i.e. 100)**<br>**CONT'D** | **B6.1.1.2.2 Count and convert Hindu Arabic numbers to Roman numerals up to 100 (C) and vice versa**<br>E.g. 1 Learners read the chart sequentially forwards and backwards, vertically (up and down), zig-zag, diagonally and randomly without pointing to the numbers. Invite 2-3 learners to read to the class. Call a Roman numeral and ask learners to write<br>E.g. 2 Give learners a numeral in the Hindu Arabic system and have learners convert to roman numeration and vice versa for instance XXIV = 24, LX = 60, XCIV = 94, CCCXXXIII = 333, etc.<br>E.g. 3 Challenge learner to find missing numbers in addition and subtraction sentences involving roman numerals. Ask learners to write similar puzzles for their friends to solve.<br>1) X + ☐ = XV<br>2) X + ☐ = XX<br>3) XXX + ☐ = XXXV<br>4) XL + ☐ = L<br>5) V + ☐ = XX<br>6) XX - ☐ = X<br>7) XXV - ☐ = XX<br>8) X - ☐ = V<br>9) XXX - ☐ = XXVII<br>10) L - ☐ = XX | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative learning; Personal Development and Leadership Attention to Precision |



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| **B6.1.1.3**<br><br>**Demonstrate understanding of factors, multiples and prime numbers from 1 to 100**<br><br>**CONT'D** | **B6.1.1.3.1 Determine the HCF and the LCM of two or three numbers using prime factors**<br><br>E.g. 1 Have learners revise the use of the factor tree method to determine prime factors of any given whole number. For example the prime factors of 24. For instance, from the figure: 24 = 2×2×2×3 so 2 and 3 are the prime factors of 24. Ask learners to list the factors of two or more given whole numbers using the factor tree; for 36 and 48 we have 36= 2×3×2×3<br>=2×2×3×3 and<br>48= 2×3×2×2×2<br>= 2×2×2×2×3.<br><br>  <br><br>E.g. 2 Learners use the prime factorization by inspection to determine the HCF by underlining the common factors in each product<br>36=**<u>2×2×3</u>**×3<br>48= 2×2×**<u>2×2×3</u>**<br><br>  <br><br>→ which is 2×2×3=12;<br><br>E.g. 3 Learners use the prime factorization by inspection to determine the LCM and underlining the largest number of factors in each product<br>36=2×2×**<u>3×3</u>**<br>48= **<u>2×2×2×2</u>**×3<br>→ which is 2×2×2×2×3×3=144<br><br>E.g. 4 Learners place factors in a Venn diagrams to find the HCF and LCM of 36 and 48. i.e. the HCF is product of factors in both circles →2×2×3=12; and the LCM is product of factors in the diagram →3×2×2×3×2×2 =144<br><br>  | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |



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| **B6.1.1.3**<br><br>**Demonstrate understanding of factors, multiples and prime numbers from 1 to 100 CONT'D** | E.g. 5 Learners use the prime factorisation to determine the LCM and HCF of three numbers using table and dividing through by prime factors. Example 1, find the LCM and HCF these sets of numbers: 18, 24 and 30. Using prime factorization table – (see figure), i.e. HCF = 2×3 = 4, = 2×2×2×3×3×5 = 360.<br><br>method in and LCM | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |

| *2 | 18 | 24 | 30 |
|---|---|---|---|
| 2 | 9 | 12 | 15 |
| 2 | 9 | 6 | 15 |
| *3 | 9 | 3 | 15 |
| 3 | 3 | 1 | 5 |
| 5 | 1 | 1 | 5 |
| | 1 | 1 | 1 |

The prime factor which can divide all the numbers is starred. The product of all the starred prime factors is the HCF. In this case the starred numbers are 2 and 3, and HCF = 6

The product of each prime factor which divided any of the numbers is the LCM



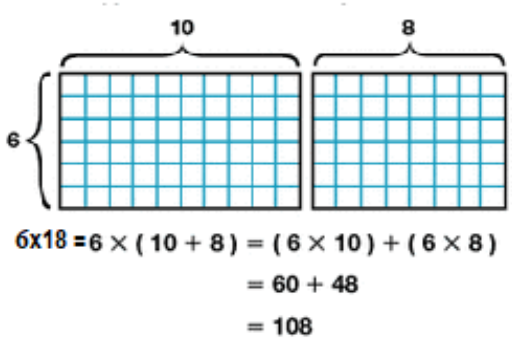
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## Sub-Strand 2: Number Operations

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| **B6.1.2.1 Describe and apply mental mathematics strategies and number properties to determine answers for basic multiplication facts to 144 and related division facts** | **B6.1.2.1.1 Apply mental mathematics strategies and number properties, such as skip counting from a known fact, using doubling or halving, using patterns in the 9s and 11s facts, using repeated doubling or halving, to determine answers for basic multiplication facts to 81 and related division facts.**<br>E.g. 1 Describe the mental Mathematics strategies used to determine a given basic fact, such as<br>– skip count up by one or two groups from a known fact (e.g., if 5 × 7 = 35, then 6 × 7 is equal to 35 + 7 and 7 × 7 is equal to 35 + 7 + 7)<br>– skip count down by one or two groups from a known fact (e.g., if 8 × 8 = 64, then 7 × 8 is equal to 64 – 8 and 6 × 8 is equal to 64 – 8 – 8)<br>– doubling (e.g., for 8 × 3 think 4 × 3 = 12, and 8 × 3 = 12 + 12)<br>– patterns when multiplying by 9 (e.g., for 9 × 6, think 10 × 6 = 60, and 60 – 6 = 54; for 7 × 9, think 7 × 10 = 70, and 70 – 7 = 63)<br>– repeated doubling (e.g., if 2 × 6 is equal to 12, then 4 × 6 is equal to 24 and 8 × 6 is equal to 48)<br>– repeated halving (e.g., for 60 ÷ 4, think 60 ÷ 2 = 30 and 30 ÷ 2 = 15)<br>– relating division to multiplication (e.g., for 64 ÷ 8, think 8 × … = 64)<br><br>E.g. 2 Recall multiplication facts to 81and related division facts | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative learning; Personal Development and Leadership<br>Attention to Precision |



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| **B6.1.2.1 Describe and apply mental mathematics strategies and number properties to determine answers for basic multiplication facts to 144 and related division facts CONT'D** | **B6.1.2.1.2 Apply mental mathematics strategies for multiplication, such as annexing then adding zero halving and doubling using the distributive property**<br>E.g. 1 Determine the products when one factor is a multiple of 10, 100, or 1000 by annexing zero or adding zeros (e.g., for 3×200 think of 3×2 and then add two zeros)<br>E.g. 2 Apply halving and doubling when determining a given product (e.g., 32×5 is the same as 16×10)<br>E.g. 3 Apply the distributive property to determine a given product involving multiplying factors that can be written in the expanded form (e.g., 6×18 = (6×10) + (6×8) = 60+48 = 108. Or apply distributive property to determine a given product involving multiplying factors that are close to multiples of 10 (e.g., 29×7 = (30×7) – (1×7) = 203.<br>E.g. 4 Explain the property for determining the answer when multiplying numbers by<br>iii. one<br>iv. zero.<br>  | Learners develop:<br>Problem Solving skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership;<br>Attention to Precision |



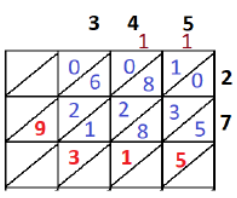
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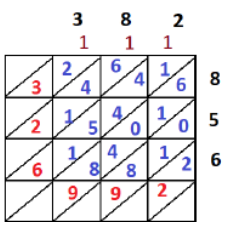
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| **B6.1.2.2**<br><br>**Demonstrate understanding of multiplication of a 2 or 3-digit number by a 2 or 3-digit number.** | **B6.1.2.2.1. Multiply multi digit numbers by 2 or 3-digit numbers efficiently**<br><br>E.g. 1. Multiplication of whole numbers using the "expand and box" method (partial decomposition method). Lead learners to multiply a 3-digit number by a 1-digit number as shown below, for example $448 \times 2 =$?<br><br>$448 \times 2 = (400 \times 2) + (40 \times 2) + (8 \times 2) = 800 + 80 + 16 = 800 + 80 + (10 + 6)$<br>$= 800 + 90 + 6 = 896$<br><br>E.g. 2. Multiplication of whole numbers using the Column or vertical method; i.e. 25×32<br><br>E.g. 3. Multiplication of whole numbers using the Distributive property; i.e. 25×32<br><br>$25 \times (32) = 25(30 + 2)$<br>$= 25(30) + 25(2)$<br>$= 750 + 50$<br>$= \mathbf{800}$<br><br>E.g. 4. Multiplication of whole numbers using the lattice method. Have learners draw a 2 by 3 lattice for solving 345 × 27; and 3 by 3 and 3 by 3 lattice for solving 382 × 856. | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |

| × | 400 | 40 | 8 |
|---|---|---|---|
| 2 | 800 | 80 | 16 |

25<br>×32<br>50<br>+750<br>**800**

25 × 30 = 750<br>25 × 2 = 50 = **800**



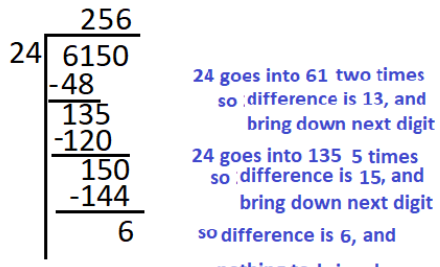



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| **B6.1.2.3**<br>**Manipulate numbers, using basic division fact up to 144** | **B6.1.2.3.1 Determine basic division fact up to 81**<br>E.g. 1. Investigate numbers that are multiples of 6, 8, 9 and 11. How do you know a number is divisible by 6, 8, 9 and 11? (This is also known as the divisibility test).<br>E.g. 2. Draw a 6 by 6 multiplication chart and use it as a game board to play the 3-in-a-line game with a pair of dice or playing cards. Players take turns in throwing a one dice and mark (or cover) a number that can be divided by the results (i.e. a quotient). The winner is the one who obtains three numbers in a line.<br>  <br>  <br>| | 5 | 6 | 7 | 8 | 9 | 10 |<br>|---|---|---|---|---|---|---|<br>| 5 | 10 | 15 | 20 | 25 | 30 | 35 |<br>| 6 | 12 | 18 | 24 | 30 | 36 | 42 |<br>| 7 | 14 | 21 | 28 | 35 | 42 | 49 |<br>| 8 | 16 | 24 | 32 | 40 | 48 | 56 |<br>| 9 | 18 | 27 | 36 | 45 | 54 | 63 |<br>| 10 | 20 | 30 | 40 | 50 | 60 | 70 | | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative learning; Personal Development and Leadership;<br>Attention to Precision |

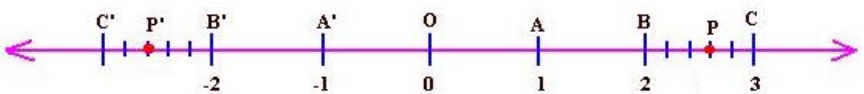


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| **B6.1.2.4**<br>**Demonstrate understanding of division of a 2 or 3-digit number by a 1 or 2-digit number** | **B6.1.2.4.1 Divide 3-digit numbers by 1-digit number efficiently**<br>E.g. 1. Division as repeated subtraction (using the long division method). Explain division as a way of repeatedly subtracting a divisor number from a given dividend until there is none left and then determining the number of times the divisor is taken from the dividend.<br>  | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership;<br>Attention to Precision |
| **B6.1.2.5**<br>**Translate word problems into mathematical sentences and solve** | **B6.1.2.5.1. Solve multi step word problems involving the four basic operations**<br>E.g. 1. Learners model mathematical statements from a given word problem involving addition and multiplication and solve using the strategies learnt<br>E.g. 2. Learners model mathematical statements from a given word problem involving division and subtraction and solve using the strategies learnt<br>E.g. 3. Learners role play a given word problem involving addition and multiplication and solve<br>E.g. 4. Learners role play a given word problem involving subtraction and division and solve<br>E.g. 5. Learners model mathematical statements from a given word problem involving division and multiplication and solve using the strategies learnt. The relationship between operations and the use of calculator and spreadsheet to assess the reasonableness of answers should be stressed. | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership;<br>Attention to Precision |

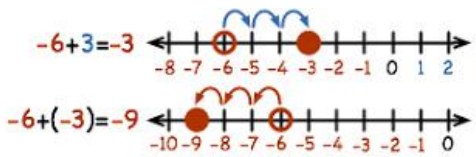


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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B6.1.2.6**<br>**Demonstrate an understanding of integers** | **B6.1.1.6.1 Locate, compare and order sets of integers using the number line and symbols "< or >"**<br>E.g. 1. Use number line to help learners to identify integers as opposites of whole numbers by answering the following questions:<br>i. Which integer is at the point marked $B^{I}$?<br>ii. Which integer is larger than $B^{I}$ and which is smaller?<br>iii. How many steps away from B is $B^{I}$?<br>  <br> | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership;<br>Attention to Precision |

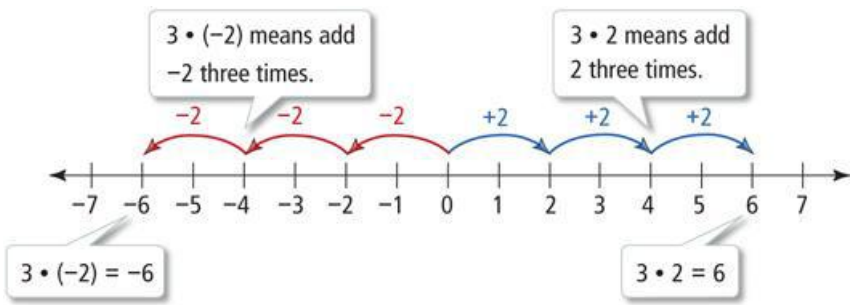


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|---|---|---|
| **B6.1.2.6**<br>**Demonstrate understanding of integers CONT'D** | **B6.1.2.6.2 Solve simple addition and subtraction problems involving integers (excluding subtraction of negative numbers)**<br>E.g. 1. Use number line to help learners to do the following types (addition)<br>(1) 9 + -4 = _____ (2) -8 + 4 = _____ (3) -3 + -5 = _____ (4) 1 + -3 = _____<br>(5) -6 + 5 = _____ (6) 6 + -2 = _____ (7) -6 + 8 = _____ (8) -2 + 9 = _____<br>  <br>E.g. 2. Learners do the following types (subtraction)<br>(9) -5 – 1 = ___ (10) -2 – 1 = ___ (11) 8 – 7) = ___ (12) 2 – 6) = _____<br>(13) -1 – 7 = ___ (14) -5 – 7) = ___ (15) -8 – 8 = ___ (16) 4 – 6 = ____<br>E.g. 3. Solve word problems; e.g. (i) Some number added to 5 is equal to -11. Find the number. (ii) Ben and Cam are diving. Ben is 15.8 meters below the surface of the water. Cam is 4.2 meters above Ben. What is Cam's position relative to the surface of the water? | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative learning; Personal Development and Leadership;<br>Attention to Precision |

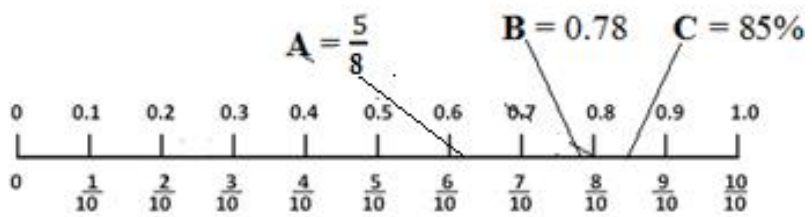


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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B6.1.2.6**<br><br>**Demonstrate an understanding of integers CONT'D** | **B6.1.1.4.3 Perform simple multiplication with integers**<br><br>E.g. 1. Use number line to help learners to multiply integers; learners should always start from zero and skip/hop to the left or write on the number line. For instance, to multiply 3 by (-2), have learners start from zero and hop to the left 3times in an interval of 2.<br><br>  <br> | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |



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| Content Standard | Indicators and EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND Core Competencies |
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| **B6.1.3.1**<br>**Demonstrate an understanding of strategies for comparing, adding, subtracting, multiplying and dividing common, decimal and percent fractions** | **B6.1.3.1.1 Compare and order a mixture of fractions: common, percent and decimal fractions (up to thousandths)**<br><br>E.g. 1. To compare and order a mixture of common, decimal and percent fractions, express them in one form (i.e. either common, decimal or percent); e.g. to order 0.758, $\frac{5}{8}$ and 73%; we have $\rightarrow \frac{5}{8} = \frac{625}{1000}$ = 62.5%, $0.758 = \frac{758}{1000}$ = 75.8%, and $73\% = \frac{73}{100}$ = 0.73, Hence the order from least to the largest is $\frac{5}{8}$, 73% and 0.758<br><br>E.g. 2. To compare and order a mixture of common, decimal and percent fractions you can locate the fractions on the number and order them<br><br>  <br> | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |



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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
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| **B6.1.3.1**<br>**Demonstrate understanding of strategies for comparing, adding, subtracting, multiplying and dividing common, decimal and percent fractions. CONT'D** | **B6.1.3.1.2 Add and subtract unlike and mixed fractions**<br>E.g. 1. To add like mixed fractions that are larger than 1, i.e. 2 1/3 and 32/3 we write down the sum of the whole numbers and add the fractions; i.e. 2 1/3 + 32/3 = 5 + 1/3 + 2/3, = 5 (1+2)/3 = 53/3= 6.<br>E.g. 2. To subtract like-fractions that are larger than 1, i.e. 2 1/3 and 32/3 , we change the mixed fractions into improper fractions and subtract;<br>i.e. 32/3 − 2 1/3 + = 11/3- 7/3 + = (11-7)/3 = 4/3. = 1 1/3<br>E.g. 3. To add or subtract improper fractions with different denominators, (2 1/3 and 32/5 ) we need find the Lowest Common Denominator (LCD) and use this to express the equivalent fractions. The LCD is 15 and the equivalent fractions are 2 5/15 and 3 6/15; their sum is 2 1/3 + 32/5 =2 5/15+ 3 6/15 = 5 (5+6)/15 which is 5 11/15; and difference 32/5 −2 1/3 = 1 (6-5)/15 = 1 1/15 | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |

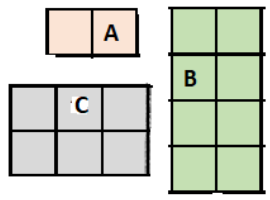


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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B6.1.3.1**<br>**Demonstrate understanding of strategies for comparing, adding, subtracting, multiplying and dividing common, decimal and percent fractions. CONT'D** | **B6.1.3.1.3 Use models to explain the result of multiplying a fraction by whole number, a whole number by a fraction and a fraction by fraction**<br><br>E.g. 1. To multiply a whole number by a mixed fraction (e.g. 3 × 2 2/3) one can multiply the whole number by the whole number and then whole number by the fraction and add the products or change the mixed fraction to improper fraction and multiply;<br><br>i.e. 3 × 2 2/3 = (3×2) + 3× 2/3) = 6 + 2/3+2/3+2/3 = 66/3 = 8 or<br><br>3 × 2 2/3 = 2 2/3+2 2/3+2 2/3 = 6 (2+2+2)/3 = 66/3 = 8<br><br>E.g. 2. To multiply a fraction by a whole number the multiplication is interpreted as "of"; e.g. 2/3 ×5 means shade 2/3 of 5 ; i.e. finding two-thirds of each of five objects; i.e. 2/3 ×5 is 2/3 of 5 quantities, which leads 10 thirds, i.e. 2/3 ×5 = 10(1/3) = 10/3 = 31/3<br><br>E.g. 3. To multiply a fraction (i.e. common or mixed) by a whole number (e.g. 4 2/5 × 5 ) first change all into common fractions, then multiply the numerators separately and multiply the denominators separately and simplify, i.e. 4 4/5 × 5 = 24/5 × 5 / ( 1 ) == (24×5)/5 = 120/5 = 24/1 = 24. [Note, the product can be simplified before multiplying the numerators separately and multiplying the denominators separately]. | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |



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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B6.1.4.1**<br><br>**Demonstrate understanding of the concept of ratios and its relationship to fractions and to the multiplication and division of whole numbers** | **B6.1.4.1.1 Use concrete models and pictorial representations to explain a ratio as a concept that shows the number of times one quantity can be obtained in another and write this symbolically and in its simplest form**<br><br>E.g. 1. Use concrete objects and/or pictorial representations to explain **ratio** as a number which tells the number of times a quantity can be obtained in another. In the figures, the area of the shape A is $\frac{1}{4}$ of the area of the shape B; so they are in the ***ratio 1:4***. Shape C is three times the size of A so the ratio of C to A is 3:1.<br>  <br>E.g. 2. Use concrete objects and/or pictorial representations to explain *simplest form of a ratio.* Shape C is made up of 6 squares and shape A is made up of 2 squares, the areas of the shapes C and A are in the ratio 6:2; and since C is three times A, the ratio 3:1 is the simplest form of 6:2. The simplest form of a ratio is obtained by dividing through by the highest common factor. The ratio of C to B is 6:8 and its simplest form is 3:4. E.g. Simplify (i) 10m : 1000km (ii) Write 4 : 12 in the form 1 : n (iii) Express 15 : 20 in the form n : 1.<br><br>E.g. 3. Solve simple problems that involve ratios and finding total ratios. E.g. (i) Out of 24 students in a class, 10 are girls. Find its simplest form the ratio of boys to girls. (ii) A boy's mass is 50kgs, and his sister's is 45kg. Find the ratio of their masses. (iii) If an orange drink is made from concentrate and water in the ratio 3:8, what fraction of the mixture is concentrate? | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |



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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B6.1.4.1 Demonstrate understanding of the concept of ratios and its relationship to fractions and to the multiplication and division of whole numbers CONT'D** | **B6.1.4.1.2 Express ratios in equivalent forms, compare and order ratios**<br><br>E.g. 1. Use the concept of ratio as a fraction to find equivalent ratios that can be easily compared. The ratio 2:3 can be expressed as 2/3 ; to determine which ratio is larger/largest change to equivalent ratios with same denominator and compare or order. E.g. Afia, Bedu and Caro each mix orange squash (S) and water (W) in the ratio 3:14, 2:7 and 1:4 respectively. Whose drink tastes strongest of squash? To determine the one Whose drink tastes strongest of squash we need to have the same unit of water, hence Bedu's.<br><br>E.g. 2. Solve simple problems that involve simplifying, comparing, finding missing and total ratios. E.g. (i) Given that 10: q = 2 : 3, find q. (ii) The ratio of boys to girls in a class room is 7 to 11. If there are a total of 49 boys in the classroom, then how many boys and girls are there altogether?<br><br>E.g. 3. Solve simple problems that involve ratios and finding total ratios. E.g. (i) Out of 24 students in a class, 10 are girls. Find its simplest form the ratio of boys to girls. (ii) A boy's mass is 50kgs, and his sister's is 45kg. Find the ratio of their masses. (iii) If an orange drink is made from concentrate and water in the ratio 3:8, what fraction of the mixture is concentrate? | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |

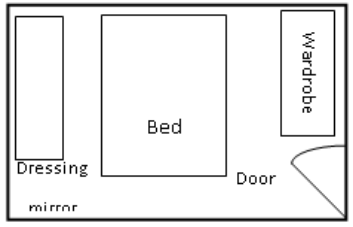


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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B6.1.4.2**<br><br>**Understand the concept of proportion and its relationship to ratios and rates. Use proportional reasoning and rates to solve real-world and mathematical problems.** | **B6.1.4.2.1 Use models to explain proportion as a comparison between quantities with equal ratios**<br><br>E.g. 1. Use mapping diagram to explain the concept of proportion as equal fractions or equivalent ratios. The mapping diagram shows that the ratio of number of goats to number of legs are equal, hence the number of goats is proportional to the number of legs. The proportion can be written as equal fractions or ratios, i.e. 3/12 = 4/16 or 3:12 = 4:16<br><br>E.g. 2. Give learners mappings to identify those that are proportional and those that are not<br><br>E.g. 3. Work out proportion in given contexts and use them in solving problems; e.g. 200 bottles of equal capacity hold 350 litres of water. How much water does each bottle hold? | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership;<br><br>Attention to Precision |



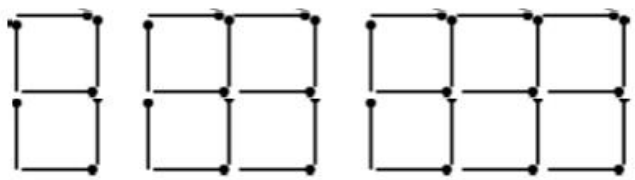
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| **B6.1.4.2**<br><br>**Understand the concept of proportion and its relationship to ratios and rates. Use proportional reasoning and rates to solve real-world and mathematical problems. CONT'D** | **B6.1.4.2.3 Use various strategies to solve proportional reasoning problems involving rates and scales**<br><br>E.g. 1. Use diagrams to explain the concept of rate a ratio that compares two different quantities measured in different units; e.g. the ratio → cost : weight = 8:2 = 4:1. This ratio is the unit ratio and show the rate for selling a unit or 1kg of the meat.<br><br>Two kilograms of meat cost ¢8. What is the cost of (i) 5kg (ii) 12kg and (iii) 25kg.<br><br>Meat (*kg*) \| 2 \| 5 \| 12 \| 25<br>Cost (*l*) \| 8 \| \| \| <br><br>E.g. 2. Work out rates and use them in solving problems. E.g. A litre of sachet water costs 40p. Find the cost of (i) $\frac{1}{2}$ litre (ii) 7 litres (iii) 9 litres<br><br>E.g. 3. Use diagrams to explain the concept of scale as a ratio that compares two different sizes – an object and its model; e.g. plan of a house or room. [Here is the plan of a bedroom. The scale is 1:100, which means that 1cm in the drawing represents 100 cm in the actual room. What is the actual length of (i) the room if it measures $4\frac{1}{2}$cm; (ii) the bed if it measures 1.8cm?]<br><br>  <br><br><br>E.g. 4. Work out scales and use them in solving problems; e.g. If 10cm on a map stands for 10km, what distances on the map stands for (i) 1km (ii) 7km (iii) 41km? | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative learning; Personal Development and Leadership<br>Attention to Precision |



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<tr>
<td>B6.2.1.1<br><br>Determine the pattern rule to make predictions about subsequent elements. CONT'D</td>
<td>
B6.2.1.1.4 Represent a given pattern visually to verify predictions<br>
E.g. 1. Ask learners to examine the pattern made with match sticks below<br>


<br>
E.g. 2. Ask learners to examine the pattern in the table below which shows the number of match sticks used in making a pattern of squares.
<table>
<tr><td>Pattern Number</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr>
<tr><td>Match sticks in the perimeter</td><td>3</td><td>6</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>Match sticks enclosed in the pattern</td><td>0</td><td>3</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
</table>
<br>
Ask learners to use match sticks to make the patterns of triangles and complete the table. Guide learners to describe the relationship<br>

</td>
<td>Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership;<br>Attention to Precision</td>
</tr>
</table>



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<tr><th>CONTENT STANDARD</th><th>INDICATORS AND EXEMPLARS</th><th>SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES</th></tr>
<tr><td></td><td>E.g. 3. Ask learners to write questions based on number or geometric patterns for their friends to find answers to; e.g. how many match sticks will be used for the 9th pattern of triangles? How many match sticks will be used for the 8th pattern of squares?<br><br><table><tr><td>Number of triangles</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td>Match sticks</td><td>7</td><td>12</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table></td><td>Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision</td></tr>
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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | CORE COMPETENCIES |
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| **B6.2.1.1**<br>**Determine the pattern rule to make predictions about subsequent elements. CONT'D** | **B6.2.1.1.7 Write a rule in words and in algebra to represent a given pattern**<br><br>E.g. 1. Ask learners to state the rules in words and in algebra to represent a given linear patterns | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |

| Term/Input | 1 | 2 | 3 | 4 | 5 | Rule for n in words | Rule for n in Algebra |
|---|---|---|---|---|---|---|---|
| Result/Output A | 9 | 18 | 27 | | | 9 times n | 9n |
| Result/Output B | 0 | 4 | 8 | 12 | | 4 times 1 less than n | 4(n-1) |
| Result/Output C | 4 | 7 | 10 | | | 1 more than 3 times n | 1 +3n |
| Result/Output D | 20 | 18 | 16 | | | 20 minus 2 times 1 less than n | 20-2(n-1) |
| Result/Output E | 15 | 19 | 23 | | | | |
| Result/Output F | 12 | 17 | 22 | | | | |
| | 100 | 85 | 70 | | | | |

**B6.2.1.1.8 Describe the relationship in a given table or chart, using a mathematical expression**

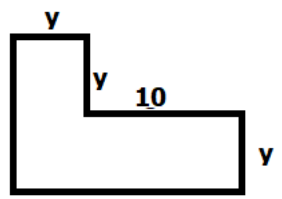
E.g.1. This table shows the pattern of cost of boxed lunches for students on a field trip. Ask learners to: (i) explain the pattern of how the cost of lunches changes as more students go on the trip; (ii) use the pattern to determine how many students went on the trip if the cost of lunches is GHC90.

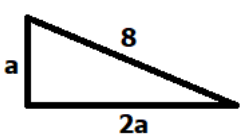
| Number of students | 1 | 2 | 3 | 4 | 5 | ? |
|---|---|---|---|---|---|---|
| Cost of lunch in Cedis | 3 | 6 | 9 | 12 | 15 | 90 |

E.g. 2. Ask learners to write questions based on the pattern for their friends to find answers



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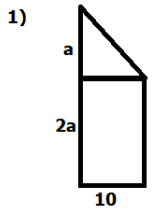
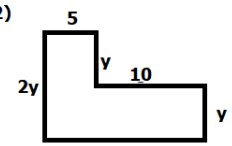
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<tr>
<td>B6.2.2.1<br><br>Demonstrate understanding of algebraic expressions</td>
<td>B6.2.2.1.1<br><br>Demonstrate understanding of algebraic expressions as mathematical phrases that can contain letters which represent ordinary numbers and operators (like add, subtract, multiply, and divide).<br><br>E.g. 1. Write algebraic expressions for word problems:<br><br>
<table>
<tr><td>1) Sum of 8 and s</td><td>2) Take away 4 from m</td></tr>
<tr><td>3) 9 times the sum of 8 and q</td><td>4) Subtract 4 from 7 times g</td></tr>
<tr><td>5) One-sixth of n is added to the product of 9 and y</td><td>6) Three-fourths of the sum of c and 2</td></tr>
<tr><td>7) 8 divided by r</td><td>8) 8 times the sum of c and 7</td></tr>
</table>
E.g. 2. Write algebraic expressions for the perimeter of the following shapes:<br><br>
1)<br> 

<br>
2)<br> 

</td>
<td>Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision</td>
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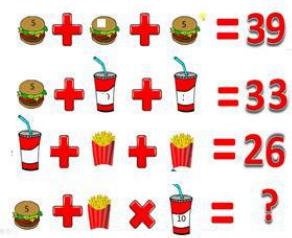


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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | CORE COMPETENCIES |
|---|---|---|
| **B6.2.2.1**<br>**Demonstrate understanding of algebraic expressions.**<br>**CONT'D** | E.g. 3. Write algebraic expressions for the area of the following shapes:<br>1)  <br>2)  <br>E.g. 4. Simplify basic algebraic expressions by grouping like terms. E.g.<br>*1)* $5 + 4z - 1 + 2z$ *2)* $10s - 1 + 3 - 5s$<br>*3)* $-6t - 7 - 2 - 3t$ *4)* $2 - z - 1 + 4z$<br>*5)* $-9c + 7c + 8 - 3c$ *6)* $-5p + 3px - 7 - 5px + 10x$<br>E.g. 5. Substitute a value for an unknown into an expression and correctly calculate the answer. E.g. Find the values of these algebraic expressions using the values given in the following:<br>1) z + 7d use z = 7 and d = 4 2) 8k + d use k = 2 and d = 3<br>3) 7(5f - 3n) - 8 use n = 3 and f = 7 4) 7d - 2f + 9 use d = 2 and f = 5<br>5) -5f + 8b + 4 - 9 use f = 9 and b = 3 6) -6(2x - 7h) use h = 2 and x = 4<br>(i)<br>(ii) If Sena used 13 match sticks in making next pattern squares,<br>(i) write a mathematical sentence that represents the matchsticks.<br>(ii) how may match sticks will she need to make the pattern with 10 squares?<br>how may squares did she make if she used 64 match sticks? | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |

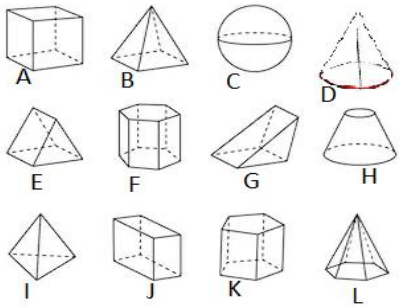


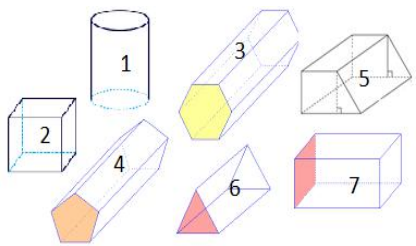
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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B6.2.3.1**<br>**Solve problems involving single-variable, one-step equations with whole number coefficients. CONT'D**<br><br>**B6.2.3.1**<br>**Solve problems involving single-variable, one-step equations with whole number coefficients. CONT'D** | **B6.2.3.1.2 Identify the unknown in a problem; represent the problem with an equation; and solve the problem concretely, pictorially or symbolically.**<br>E.g. 1 Learners find the value of *x* in equations.<br>8. $4 + x = 11$<br>9. $x - 3 = 10$<br>10. $4x = 12$<br>11. $24 = 3x$<br>**12.** $\frac{x}{2} = 5$<br>**B6.2.3.1.3 Create a problem for a given equation.**<br>E.g. 1 Ask students to describe two different stories that the equation $4 - x = 9$ could represent.<br>E.g. 2. Ask learners to solve puzzle in the figure, by solving the equations in each line;<br>  | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |



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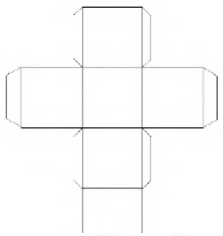
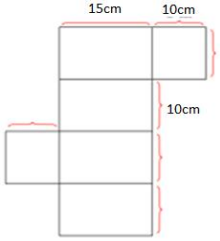
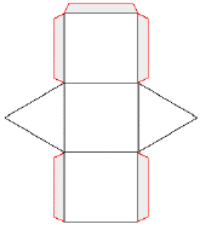
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<tr><th>CONTENT STANDARD</th><th>INDICATORS AND EXEMPLARS</th><th>SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES</th></tr>
<tr>
<td>B6.3.1.1<br><br>Demonstrate understanding of prisms</td>
<td>
B6.3.1.1.1 Identify examples of rectangular and triangular prisms in the classroom and the community<br><br>
E.g. 1 Give learners models of 3-D shapes made from cardboard to examine and describe their cross sections (i.e. the surface or shape exposed by making a straight cut through something, especially at right angles to an axis). Ask learners to draw and complete the table for the description of the cross sections of the 3-D objects<br><br>


<br><br>
E.g. 2 Ask learners to sort 3-D shapes with uniform cross sections; these are prisms and they are named after their uniform cross sections. Ask them to identify and name the following prisms<br><br>
<table>
<tr><th></th><th></th><th colspan="2">Cross Section</th></tr>
<tr><th></th><th></th><th>Shape</th><th>Uniform/Not-Uniform</th></tr>
<tr><td>A.</td><td>Cube</td><td>square</td><td>uniform</td></tr>
<tr><td>B.</td><td>Pyramid</td><td>rectangle</td><td>not uniform</td></tr>
<tr><td>C.</td><td></td><td></td><td></td></tr>
<tr><td>D.</td><td></td><td></td><td></td></tr>
<tr><td>E.</td><td></td><td></td><td></td></tr>
<tr><td>F.</td><td></td><td></td><td></td></tr>
</table>
<br>


</td>
<td>Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision</td>
</tr>
</table>

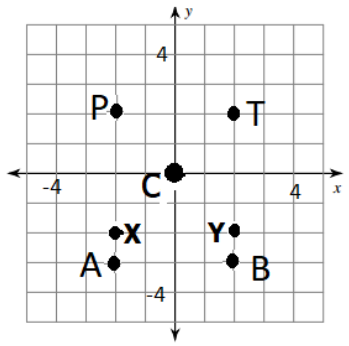
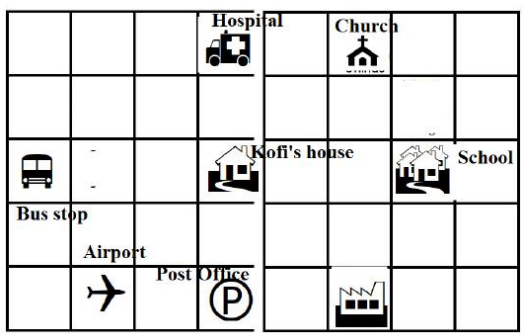


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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B6.3.1.1**<br>**Demonstrate an understanding of prisms. CONT'D** | **B6.3.1.1.2 Construct rectangular and triangular prisms from their nets.**<br>E.g. 1 Give the learners rectangular and triangular prisms made from cardboard by yourself (or cardboard packing cases from shops) to open and examine their nets<br>  <br>E.g. 2 Give the learners some cardboard or manila card to draw a net of a square prism whose cross section is a square with sides 10cm by 10cm and height 10cm. After drawing add tabs so that it can be folded so that the tabs are glued inside<br>  <br>E.g. 3 Give the learners some cardboard or manila card to draw a net of a rectangular prism whose cross section is a square with sides 10cm by 10cm and height 15cm. After drawing add tabs so that it can be folded so that the tabs are glued inside<br>  <br>E.g. 4 Give the learners some cardboard or manila card to draw a net of a triangular prism whose cross section is an equilateral triangle with side 10cm and height 10 cm. After drawing add tabs so that it can be folded so that the tabs are glued inside<br>  | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |

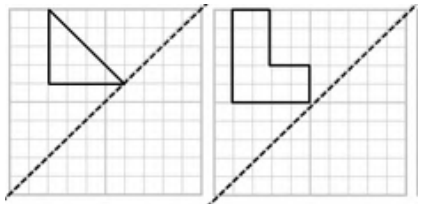
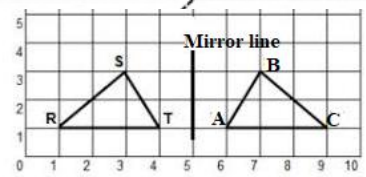
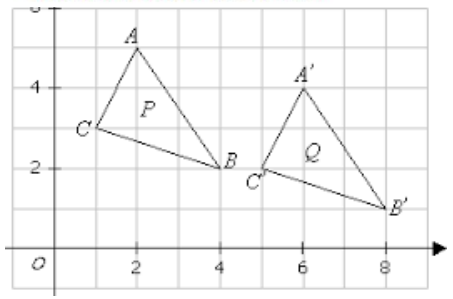


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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B6.3.3.5**<br>**Describe the position of objects in space using the cardinal points** | **B6.3.3.5.1 Tell the position and motion of objects in space using the cardinal points north-east, north-west, south-east and south-west**<br>E.g. 1. Pupils describe the position and motion of objects from one point to another using the 4 main cardinal points and the half-betweens: e.g. The point **P** is north of the point **P** X but north-west of point C. Y is south of T but south-west of C.<br>  <br>E.g. 2. Pupils describe the position of places in a grid from a given location. E.g. the church is north-east from Kofi's house; the hospital is north-west from Kofi's house.<br>E.g. 3. Ask learners to give directions from Kofi's house the various locations. E.g. Move 2 squares to the east and 2 squares to the north to get to the church.<br>  | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |



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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B6.3.3.5**<br><br>**Perform a single transformation (i.e. reflection translation) on a 2D shape** | **B6.3.3.5.1 Perform and identify images of a single transformation (i.e. reflection) on 2D shapes in a plane**<br><br>E.g. 1. Give learners photocopied worksheets with several incomplete 2D shapes to review the drawing of images of objects when reflected in line of symmetry which is outside the object<br><br>E.g. 2. Review the properties of images under reflection, i.e. object and image sizes and shape, distances from mirror line and shape<br><br>E.g. 3 Use a grid or graph board to demonstrate the 'horizontal followed by vertical' movements called translation. E.g. In the figure is *'the movement which takes point four units horizontally to the right and one unit vertically down' or 4 right, one down',* is a vector or translation vector (2→, 1↓) and written as $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$. Q is the image of P under the translation by the vector $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$<br><br>  <br>the<br><br>  <br><br>△RST is the image of △ABC in the reflection in the in the mirror line<br>*each*<br><br>  <br> | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |

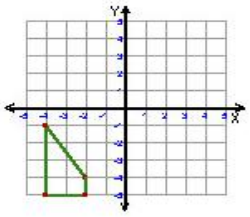
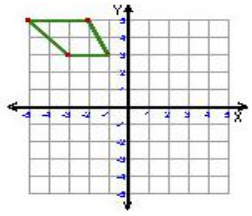
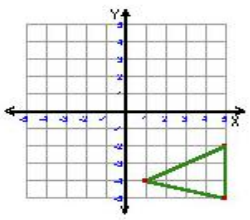
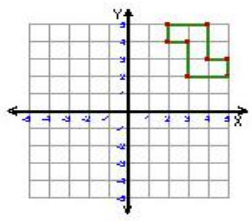
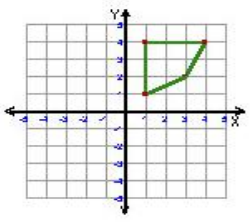
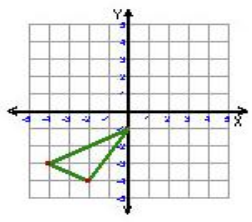


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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B6.3.3.5**<br>**Perform a single transformation (i.e. reflection translation) on a 2D shape. CONT'D** | E.g. 4 Discuss the properties of images under translations, i.e. object and image sizes and shape<br>E.g. 5 Give learners photocopied worksheets with several 2D shapes and ask them to draw images of each under the translation vector given<br>**Note: Diagram is on the next page** | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |

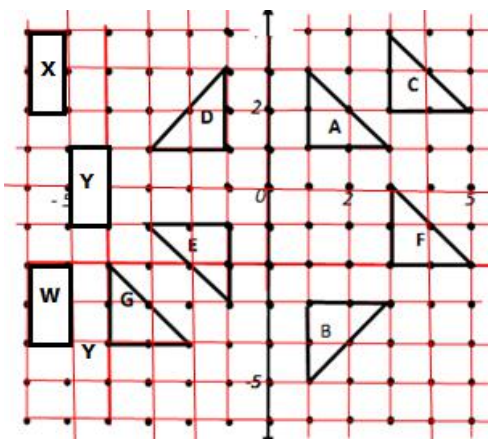


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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | CORE COMPETENCIES |
|---|---|---|
| **B6.3.3.5**<br>**Perform a single transformation (i.e. reflection translation) on a 2D shape.**<br>**CONT'D** | 1) Translation: 3 right<br>  <br>2) Translation: 2 right and 3 down<br>  <br>3) Translation: 5 left<br>  <br>4) Translation: 4 left and 3 down<br>  <br>5) Translation: 4 left and 5 down<br>  <br>6) Translation: 3 right and 4 up<br>  | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |

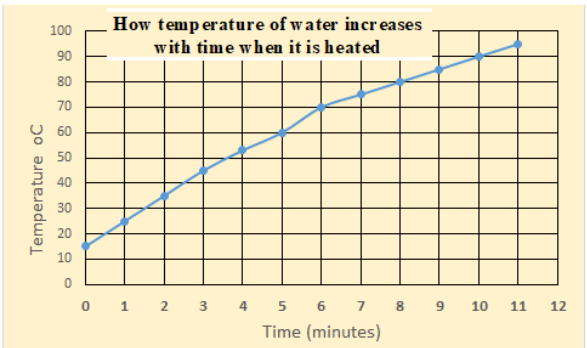


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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | CORE COMPETENCIES |
|---|---|---|
| **B6.3.3.5**<br>**Perform a single transformation (i.e. reflection translation) on a 2D shape.**<br>**CONT'D** | **B6.3.3.5.2 Identify images of a single transformation (i.e. reflection and translation) of a 2D shapes in a plane**<br>E.g. 1. Ask learners to identify images of reflections of 2D shapes in the figure. Ask learners to study the shapes, identify their images. E.g. shape D is a reflection of shape E in the line y = 0 or x-axis; shape A is a reflection of shape B in the line $y = -1$; and Shape X is a reflection of shape W in the line $y = 0$<br>E.g. 2. Ask learners to identify images of translations of given 2D shapes in a given plane. Ask learners to study the shapes, identify their images and the translation vectors. E.g. shape F is a translation of shape A by the translation vector $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$ (or 2 right, 3 three down); and Shape X is a translation of shape W by the translation vector $\begin{pmatrix} 0 \\ 6 \end{pmatrix}$ (or 0 right, 6 up)<br>  | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |



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<table>
<tr><th>CONTENT STANDARD</th><th>INDICATORS AND EXEMPLARS</th><th>SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES</th></tr>
<tr>
<td>B6.4.1.1<br>Create, label, and interpret line graphs to draw conclusions.<br>CONT'D</td>
<td>
B6.4.1.1.1 Draw a line graph by determining the common attributes (title, axes and intervals) and draw the graph for a given table of values
<table>
<tr><td>Time (minutes)</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td></tr>
<tr><td>Temperature (°C)</td><td>15</td><td>25</td><td>35</td><td>45</td><td>53</td><td>60</td><td>70</td><td>75</td><td>80</td><td>85</td><td>90</td><td>95</td></tr>
</table>
E.g. 1. Give learners the data presented in tables and ask them to determine the common attributes (title, axes and intervals) and draw a line graph. Let them understand line graphs are used when we want to show how something changes over time in relation to something else<br><br>
E.g. 2. Ask learners to draw a line graph by determining the common attributes (title, axes and intervals) and completing the drawing with title and labelled axes for the data below on how temperature of water increases with time when it is heated.<br><br>
E.g. 3. Ask learners to study the line graph and answer questions based on them. For instance, after what time will the water temperature reach 70°C? What is the water temperature after 5 minutes?<br>


</td>
<td>Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision</td>
</tr>
</table>



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<table>
<tr><th>CONTENT STANDARD</th><th>INDICATORS AND EXEMPLARS</th><th>SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES</th></tr>
<tr><td>B6.4.1.1<br>Create, label, and interpret line graphs to draw conclusions.<br>CONT'D</td><td>
B6.4.1.1.2 Determine whether a given set of data can be represented by a line graph (continuous data) or a series of points (discrete data) and explain why<br>
E.g. 1. Through discussion, guide learners to understand the difference between discrete and continuous data. Ask them to sort the following tables of values into the two categories – those with discrete data and those with continuous data<br>
[Discrete data take particular values, while continuous data are not restricted to separate values. Discrete data can take on only integer values whereas continuous data can take on any value. For instance the number of malaria patients treated by a hospital each year is discrete but your weight recorded of over a year is continuous]<br>
(i) The number of match sticks Sena used in making a pattern of squares
<table>
<tr><td>Number of Squares (n)</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr>
<tr><td>Matchsticks (s)</td><td>4</td><td>7</td><td>10</td><td>13</td><td>16</td><td>19</td><td>22</td><td>25</td></tr>
</table>
(ii) Progress of plant's growth recorded by Kofi over a week
<table>
<tr><td>Days (d)</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr>
<tr><td>Height (h) (cm)</td><td>5</td><td>6.5</td><td>8</td><td>9.5</td><td>11</td><td>12.5</td><td>14</td><td>15.5</td></tr>
</table>
(iii) Number of people in the immediate family of P6 students in a school
<table>
<tr><td>Number of people in family</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr>
<tr><td>Number of students</td><td>0</td><td>4</td><td>9</td><td>11</td><td>6</td><td>5</td><td>3</td><td>3</td><td>1</td></tr>
</table>
(iv) Abu travelled on a bicycle to buy food, the table shows how far (in kilometres) he was away from the house.
<table>
<tr><td>Time (minutes)</td><td>0</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td><td>30</td></tr>
<tr><td>Distance (km)</td><td>0</td><td>3</td><td>3.5</td><td>4</td><td>4</td><td>2</td><td>0</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
</table>
E.g. 2. Ask learners to write questions for their friends to read and interpret the tables
</td><td>Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision</td></tr>
</table>



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<table>
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<tr>
<td>B6.4.1.1<br><br>Create, label, and interpret line graphs to draw conclusions.<br><br>CONT'D</td>
<td>
B6.4.1.1.3 Apply understanding of how to create a line graph by using a given table of values (or set of data) to draw a line graph and answer questions based on them to interpret and draw conclusions<br><br>
E.g. 1. Ask learners to draw more line graphs for given data values by determining the common attributes (title, axes and intervals) and pose, as well as answer, questions based on them to interpret the line graphs to draw conclusions<br><br>
(i) Progress of plant's growth recorded by Kofi over a week
<table>
<tr><td>Days (d)</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr>
<tr><td>Height (h) (cm)</td><td>5</td><td>6.5</td><td>8</td><td>9.5</td><td>11</td><td>12.5</td><td>14</td><td>15.5</td></tr>
</table>
(ii) Cost of boxed lunches for students on a field trip.
<table>
<tr><td>Number of students</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr>
<tr><td>Cost of lunch (¢)</td><td>3</td><td>6</td><td>9</td><td>12</td><td>15</td><td>18</td><td>21</td><td>24</td></tr>
</table>
(iii) Abu travelled on a bicycle to buy food, the table shows how far (in kilometers) he was away from the house.
<table>
<tr><td>Time (minutes)</td><td>0</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td><td>30</td></tr>
<tr><td>Distance (km)</td><td>0</td><td>3</td><td>3.5</td><td>4</td><td>4</td><td>2</td><td>0</td></tr>
</table>
(iv) Average monthly rainfall in mm recorded for two towns in a year
<table>
<tr><th></th><th>J</th><th>F</th><th>M</th><th>A</th><th>M</th><th>J</th><th>J</th><th>A</th><th>S</th><th>O</th><th>N</th><th>D</th></tr>
<tr><td>Kumasi</td><td>5</td><td>10</td><td>15</td><td>20</td><td>50</td><td>45</td><td>55</td><td>35</td><td>40</td><td>50</td><td>35</td><td>10</td></tr>
<tr><td>Oda</td><td>3</td><td>10</td><td>13</td><td>25</td><td>40</td><td>50</td><td>60</td><td>50</td><td>40</td><td>45</td><td>35</td><td>8</td></tr>
</table>
E.g. 3. Ask learners to study the line graph and answer questions based on them. For instance, after what time will the water temperature reach 70oC? What is the water temperature after 5 minutes?<br><br>
E.g. 3. Ask learners to write questions for their friends to read and interpret the line graphs
</td>
<td>Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision</td>
</tr>
</table>



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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B6.4.1.2**<br><br>**Select, justify, and use appropriate methods of collecting data, including questionnaires, interview, observation, experiments, databases, electronic media, etc.** | **B6.4.1.2.1 Select a method for collecting data to answer a given question and justify the choice**<br><br>E.g. 1. Lead learners (in small groups) to discuss and write down how they would make decisions in the following situations, what facts they would take into account and how they would collect these 'facts'.<br>(a) The type of drinks to buy for a class party<br>(b) The make of sport shoes to buy for all P6 students<br>(c) The make of school bag to buy for all P6 students<br>(d) The number of desks in each classroom<br>(e) How much money P6 students spend on bus fare to school every month?<br>(f) To buy drinks for people in the immediate family of all P6 students at a party.<br>(g) Buy a mobile phone from an online shop<br><br>E.g. 2. Lead a discussion on the methods of data collection below and ask them to identify which method they will use to gather the facts for each situation (i.e. in E.g. 1. above)<br>• questionnaires<br>• interview<br>• observation<br>• experiments<br>• databases<br>• electronic media or internet | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |



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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B6.4.1.2**<br>**Select, justify, and use appropriate methods of collecting data, including questionnaires, interview, observation, experiments, databases, electronic media, etc.**<br>**CONT'D** | **B6.5.2.2 Design and administer a questionnaire for collecting data to answer a given question and record the results**<br>E.g. 1. Guide learners (in small groups) to do surveys; as surveys are useful ways of getting them to interact, produce question forms and collect real information. Ask them to use Question Form below for collecting the class data<br>**Class Survey Question Form**<br>1) Hello, What is your name? _____<br>2) How old are you? _____<br>3) What is your favourite school subject? _____<br>4) What is your worst subject? _____<br>5) What is the most important school subject? _____<br>6) The size of your shoe? _____<br>7) What is your favourite drink? _____<br>8) How much do you spend on bus fare to school every day? _____ | Learners develop:<br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |



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<table>
<tr><th>CONTENT STANDARD</th><th>INDICATORS AND EXEMPLARS</th><th>SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES</th></tr>
<tr><td>B6.4.1.2<br>Select, justify, and use appropriate methods of collecting data, including questionnaires, interviews, observations, experiments, databases, electronic media, etc.<br>CONT'D</td><td>E.g. 2. Guide learners in each group to use a table (like the one below) to organize the data obtained with the question form<br><br>
<table>
<tr><th>Name</th><th>Age</th><th>Favourite subject</th><th>Worst subject</th><th>Important subject</th><th>Shoe size</th><th>Favourite mineral</th><th>Daily bus fare (cedis)</th></tr>
<tr><td>Kojo</td><td>12</td><td>English</td><td>Maths</td><td>English</td><td>6</td><td>Coke</td><td>2</td></tr>
<tr><td>Aku</td><td>11</td><td>Science</td><td>P.E.</td><td>English</td><td>6</td><td>Coke</td><td>1</td></tr>
<tr><td>Ami</td><td>11</td><td>Maths</td><td>Art</td><td>Maths</td><td>6</td><td>Fanta</td><td>3</td></tr>
<tr><td>Abu</td><td>13</td><td>Maths</td><td>Art</td><td>Maths</td><td>7</td><td>Sprite</td><td>1</td></tr>
<tr><td>Ama</td><td>12</td><td>Science</td><td>Art</td><td>Maths</td><td>7</td><td>Fanta</td><td>2</td></tr>
<tr><td>Paapa</td><td>11</td><td>Maths</td><td>P.E.</td><td>Science</td><td>6</td><td>Fanta</td><td>1</td></tr>
</table></td><td>Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision</td></tr>
</table>

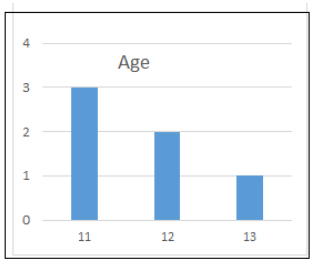


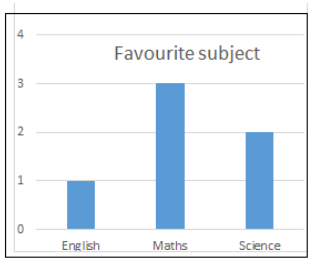
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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B6.4.1.2**<br><br>**Select, justify, and use appropriate methods of collecting data, including questionnaires, interview, observation, experiments, databases, electronic media, etc. CONT'D** | **B6.4.1.2.3 Design and administer a questionnaire for collecting data to answer given question(s), record the data, analyse and graph the results to solve problems**<br><br>E.g. 1. Guide learners (in each small group) to use tallies and count the results obtained in the organized data table (see P6.5.2.2.3 above) to do a frequency table for each question.<br><br>E.g. 2. Ask learners to study the results in the frequency tables to draw bar or line graphs and write their conclusion.<br><br>E.g. 3. Ask learners to identify questions for a survey, do a question form and collect real data, organize the data obtained into frequency tables and graph it using bar or line graphs<br><br>and the results to solve problems. They may do surveys on the following situations or issues:<br><br>(a) The type of drinks to buy for a class party.<br>(b) The make of sport shoes to buy for all P6 students.<br>(c) The make of school bag to buy for all P6 students. | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |

| Age | Frequency |
|---|---|
| 11 | 3 |
| 12 | 2 |
| 13 | 1 |
| Total | 6 |

| Favourite subject | Frequency |
|---|---|
| English | 1 |
| Maths | 3 |
| Science | 2 |
| Total | 6 |






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<table>
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<tr>
<td>B6.4.2.2<br>Demonstrate an understanding of probability by identifying all possible outcomes of a probability experiment, determining the theoretical and experimental probability of outcomes in a probability experiment</td>
<td>
B6.4.2.2.1 List the possible outcomes of a probability experiment, such as tossing a coin, rolling a die with a given number of sides, spinning a spinner with a given number of sectors and determine the theoretical probability of an outcome occurring for a given probability experiment<br><br>
E.g. 1. Through discussion guide learners to understand that theoretical probability is what we expect to happen, where experimental probability is what actually happens when we try it out. The both probabilities are calculated the same way, using the number of possible ways an outcome can occur divided by the total number of outcomes<br><br>
E.g. 2. Guide learners (in each small group) to carry out the following experiments 100 times, use tallies to record their results, and transfer it to frequency tables: (i) tossing a coin 100 times; (ii) throwing a dice 100 times<br><br>
Tossing a coin 100 times
<table>
<tr><th>Outcome</th><th>Number of throws</th></tr>
<tr><td>Head</td><td>42</td></tr>
<tr><td>Tail</td><td>58</td></tr>
</table>

Throwing a dice 100 times
<table>
<tr><th>Outcome</th><th>Number of throws</th></tr>
<tr><td>1</td><td>18</td></tr>
<tr><td>2</td><td>23</td></tr>
<tr><td>3</td><td>21</td></tr>
<tr><td>4</td><td>15</td></tr>
<tr><td>5</td><td>13</td></tr>
<tr><td>6</td><td>10</td></tr>
</table>
</td>
<td>Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision</td>
</tr>
</table>



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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
|---|---|---|
| **B6.4.2.2**<br>**Demonstrate an understanding of probability by identifying all possible outcomes of a probability experiment, determining the theoretical and experimental probability of outcomes in a probability experiment.**<br><br>**CONT'D** | E.g. 3 Guide learners (in each small group) to use the results of the experiments above (recorded in the tables above) to work out the experimental probability and compare to the theoretical probability. E.g. the experiment probability of a head showing up out of the hundred outcomes is given by $\frac{42}{100}$ =0.42; and the theatrical probability is $\frac{50}{100}$ = 0.5.<br><br>E.g. 4 Ask learners (in each small group) use the results of the experiments above (recorded in the tables above) to work out the experimental probability and compare to the theoretical probability of the following events (i.e. second table)<br>i. rolling a 2<br>ii. rolling a number greater than 4<br>iii. rolling a 1 or a 3<br><br>  <br><br>E.g. 5 Put the results from all the small groups together ask the class to work out the experimental probabilities and compare to the theoretical probabilities of the events<br>i. rolling a 2<br>ii. rolling a number greater than 4<br>iii. rolling a 1 or a 3<br><br>  <br><br>E.g. 6 Ask the learners their observations on whether or not the experimental probability is getting closer to the theoretical probability | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |



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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
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| **B6.4.2.2**<br>**Demonstrate an understanding of probability by identifying all possible outcomes of a probability experiment, determining the theoretical and experimental probability of outcomes in a probability experiment. CONT'D.** | **B6.4.2.2.2 Predict the probability of a given outcome occurring for a given probability experiment by using theoretical probability**<br><br>E.g. 1. Guide learners (in each small group) to carry out the following experiments 100 times, use tallies to record their results, and transfer it to frequency tables: (i) spinning a 5-sector spinner 100 times; (ii) spinning an 8-sector spinner 100 times<br><br>8-sector spinner<br>Outcome / Number of throws<br>1 / 15<br>2 / 16<br>3 / 15<br>4 / 15<br>5 / 13<br>6 / 10<br>7 / 7<br>8 / 9<br><br>5-sector spinner<br>Outcome / Number of throws<br>1 / 18<br>2 / 23<br>3 / 25<br>4 / 16<br>5 / 18<br><br>  <br>  <br><br>E.g. 2 Ask learners (in each small group) to use the results of the experiments above (recorded in the tables above) to work out the experimental probability and compare to the theoretical probability of the following events with each of the spinners (i.e. table above)<br>i. pinning a 2<br>ii. pinning a number greater than 4<br>iii. pinning a 1 or a 3 | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership Attention to Precision |



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| CONTENT STANDARD | INDICATORS AND EXEMPLARS | SUBJECT SPECIFIC PRACTICES AND CORE COMPETENCIES |
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| **B6.4.2.2**<br>**Demonstrate an understanding of probability by identifying all possible outcomes of a probability experiment, determining the theoretical and experimental probability of outcomes in a probability experiment**<br><br>**CONT'D** | **P6.4.2.2.3 Explain that the experimental probability approaches the theoretical probability of a particular outcome as the number of trials in an experiment increases**<br><br>E.g. 1 Put the results from all the small groups for (a) spinning the 5-sector spinner, (b) spinning the 8-sector spinner, together. Ask the class to work out the experimental probabilities and compare to the theoretical probabilities of the events<br>i. pinning a 2<br>ii. pinning a number greater than 4<br>iii. pinning a 1 or a 3<br><br>E.g. 2 Ask learners their observations as to whether or not the experimental probability is getting closer to the theoretical probability. Ask them to explain the difference between theoretical probability and experimental probability | Learners develop:<br><br>Problem Solving Skills; Critical Thinking; Justification of Ideas; Collaborative Learning; Personal Development and Leadership<br><br>Attention to Precision |



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